

1 Basic Concepts of Continuous Optimization

In this section, we explain the basic concepts of continuous optimization, including convexity, strong convexity, smoothness, and positive definiteness of the Hessian.

1.1 Convexity and Strong Convexity

The notions of convexity and strong convexity are fundamental in optimization theory. For simplicity, let us assume that f is a real-valued function defined over the entire space $\mathbb{R}^n$, i.e., $f: \mathbb{R}^n \rightarrow \mathbb{R}$. Also, let $\mu > 0$ be a constant. The function f is convex or μ -strongly convex if and only if the following hold for any $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$:

$$(\text{convex}) \quad f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y),$$

$$(\mu\text{-strongly convex}) \quad f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y) - \frac{\mu}{2}\lambda(1-\lambda)\|x-y\|^2.$$

There are more intuitive equivalent definitions of convexity and strong convexity using the gradient of f . Assume that f is of class C^1 and it takes real values over the domain $\mathbb{R}^n$. The function f is convex or μ -strongly convex if and only if the following hold for any $x, y \in \mathbb{R}^n$:

$$(\text{convex}) \quad f(y) \geq f(x) + \nabla f(x)^\top (y-x),$$

$$(\mu\text{-strongly convex}) \quad f(y) \geq f(x) + \nabla f(x)^\top (y-x) + \frac{\mu}{2}\|y-x\|^2.$$

We show the equivalence of these two definitions below.

Proposition 1. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^1 . Then the two definitions of convexity and μ -strong convexity are equivalent, respectively.*

Proof. We show the case of μ -strong convexity. First, assume that the former definition holds. By the definition of the gradient and the C^1 assumption on f , we have

$$\begin{aligned} \nabla f(x)^\top (y-x) &= \lim_{\lambda \rightarrow 0} \frac{f(x + \lambda(y-x)) - f(x)}{\lambda} \\ &\leq \lim_{\lambda \rightarrow 0} \frac{1}{\lambda} \left((1-\lambda)f(x) + \lambda f(y) - \frac{\mu}{2}\lambda(1-\lambda)\|x-y\|^2 - f(x) \right) \\ &= f(y) - f(x) - \frac{\mu}{2}\|x-y\|^2. \end{aligned}$$

By rearranging the terms appropriately, we can show that the latter definition holds.

Conversely, assume that the latter definition holds. Let $z := (1-\lambda)x + \lambda y$. Then we have

$$f(x) \geq f(z) + \nabla f(z)^\top (x-z) + \frac{\mu}{2}\|x-z\|^2,$$

$$f(y) \geq f(z) + \nabla f(z)^\top (y-z) + \frac{\mu}{2}\|y-z\|^2.$$

Multiplying the first inequality by $1-\lambda$ and the second by λ , and adding them together, we deduce the following. Note that the terms involving $\nabla f(z)$ cancel out entirely by the definition of z .

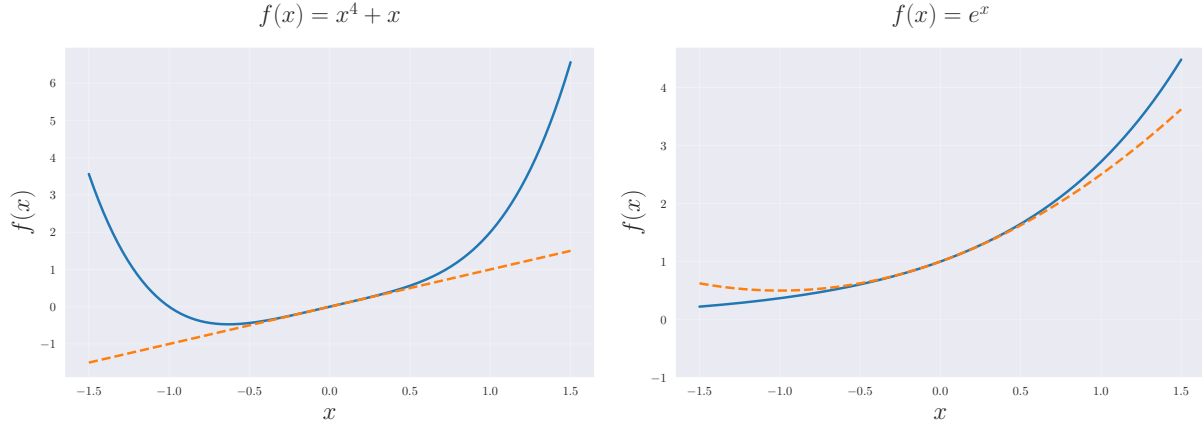
$$\begin{aligned} &(1-\lambda)f(x) + \lambda f(y) \\ &\geq f(z) + \nabla f(z)^\top ((1-\lambda)(x-z) + \lambda(y-z)) + \frac{\mu}{2} \left((1-\lambda)\|x-z\|^2 + \lambda\|y-z\|^2 \right) \\ &= f(z) + \frac{\mu}{2} \left((1-\lambda)\lambda^2\|x-y\|^2 + \lambda(1-\lambda)^2\|y-x\|^2 \right) \\ &= f(z) + \frac{\mu}{2}\lambda(1-\lambda)\|x-y\|^2. \end{aligned}$$

Thus, the former definition holds. This shows that these definitions are equivalent.

By considering the case of $\mu = 0$, the convex case can be shown in the same way. □

Strong convexity indicates that, in addition to convexity, the objective function has uniformly positive curvature. We show examples of convex and strongly convex functions in Figure 1. Such definitions may vary slightly in the literature. For example, see [2].

Convex but not strongly convex



Strongly convex

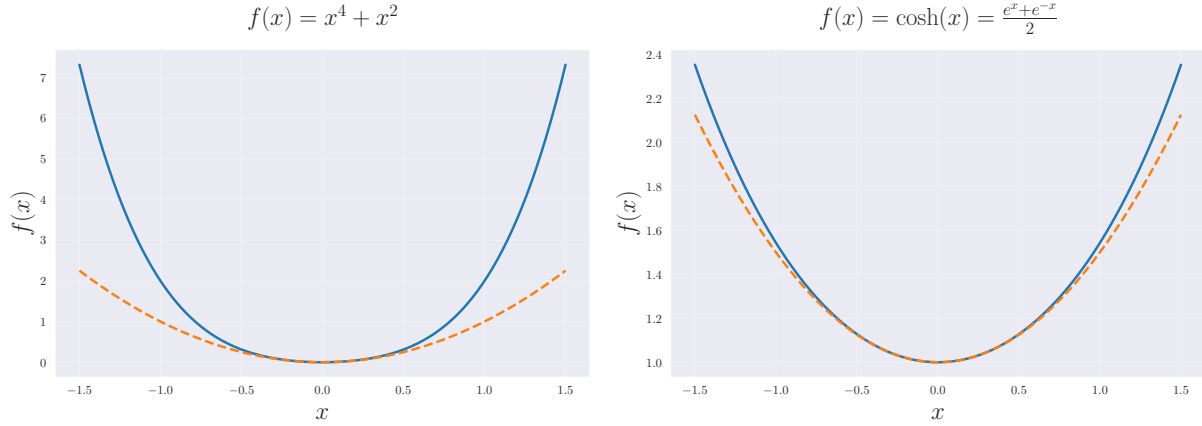


Figure 1: Examples of convex and strongly convex functions. The dashed line shows the quadratic approximation at $x = 0$. The upper two functions are convex but not strongly convex. The lower two functions are strongly convex, as there exists $\mu > 0$ satisfying the definition of strong convexity.

1.2 Positive Definiteness of the Hessian

Next, assuming that f is of class C^2 , we show how convexity and strong convexity relate to the definiteness of the Hessian matrix $\nabla^2 f(x)$.

We define the definiteness of matrices beforehand. Let A be a symmetric matrix in $\mathbb{R}^{n \times n}$. A matrix A is called positive or negative definite (or semi-definite) based on the following conditions:

- (positive definite) $v^\top A v > 0 \quad \forall v \in \mathbb{R}^n \setminus \{0\},$
- (positive semi-definite) $v^\top A v \geq 0 \quad \forall v \in \mathbb{R}^n,$
- (negative definite) $v^\top A v < 0 \quad \forall v \in \mathbb{R}^n \setminus \{0\},$
- (negative semi-definite) $v^\top A v \leq 0 \quad \forall v \in \mathbb{R}^n.$

A matrix is indefinite if it is neither positive nor negative definite. For matrices $A, B \in \mathbb{R}^{n \times n}$, the notation $A \succeq B$ indicates that $A - B$ is positive semi-definite. In particular, if B is a zero matrix, we may simply write $A \succeq 0$. We similarly define $\preceq$ for negative semi-definiteness. For $\mu \geq 0$, the condition $A \succeq \mu I$ is equivalent to $v^\top A v \geq \mu \|v\|^2$ for

all $v \in \mathbb{R}^n$. It directly implies that all eigenvalues of A are at least μ . It also implies that the operator norm $\|A\|$ is at least μ , since the operator norm of a symmetric matrix is equal to an absolute value of its largest eigenvalue.

Here, strong convexity means that the function has uniformly positive curvature. This corresponds to the Hessian matrix also having uniformly positive curvature, i.e., being uniformly positive definite. In fact, it is known that convexity and strong convexity are equivalent to the definiteness of the Hessian matrix as follows.

Proposition 2. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^2 . Then*

- *f is convex if and only if $\nabla^2 f(x) \succeq 0$ holds for all $x \in \mathbb{R}^n$.*
- *f is μ -strongly convex if and only if $\nabla^2 f(x) \succeq \mu I$ holds for all $x \in \mathbb{R}^n$.*

Proof. Firstly, assume that f is μ -strongly convex. For any $x \in \mathbb{R}^n$, $v \in \mathbb{R}^n$ and $t > 0$, letting $y = x \pm tv$ gives

$$\begin{cases} f(x+tv) \geq f(x) + t\nabla f(x)^\top v + \frac{\mu}{2}t^2\|v\|^2, \\ f(x-tv) \geq f(x) - t\nabla f(x)^\top v + \frac{\mu}{2}t^2\|v\|^2. \end{cases}$$

By Taylor's theorem, there exists $s_\pm \in (0, 1)$ such that

$$\begin{cases} f(x+tv) = f(x) + t\nabla f(x)^\top v + \frac{1}{2}t^2v^\top \nabla^2 f(x+s_+tv)v, \\ f(x-tv) = f(x) - t\nabla f(x)^\top v + \frac{1}{2}t^2v^\top \nabla^2 f(x-s_-tv)v. \end{cases}$$

Combining these results gives

$$v^\top \frac{\nabla^2 f(x+s_+tv) + \nabla^2 f(x-s_-tv)}{2} v \geq \mu\|v\|^2.$$

Letting $t \rightarrow 0$ and using the continuity of $\nabla^2 f$ from the C^2 assumption gives

$$v^\top \nabla^2 f(x) v \geq \mu\|v\|^2.$$

Because $v \in \mathbb{R}^n$ is arbitrary, we obtain $\nabla^2 f(x) \succeq \mu I$.

Conversely, let $\mu > 0$ and assume $\nabla^2 f(x) \succeq \mu I$ for all $x \in \mathbb{R}^n$. By Taylor's theorem, for any $x, y \in \mathbb{R}^n$, letting $\phi(t) := f(x + t(y-x))$ gives

$$\begin{aligned} f(y) &= \phi(1) \\ &= \phi(0) + \phi'(0) + \int_0^1 (1-t)\phi''(t)dt \\ &= f(x) + \nabla f(x)^\top (y-x) + \int_0^1 (1-t)(y-x)^\top \nabla^2 f(x+t(y-x))(y-x)dt. \end{aligned}$$

By the assumption $\nabla^2 f(x) \succeq \mu I$, we have

$$\int_0^1 (1-t)(y-x)^\top \nabla^2 f(x+t(y-x))(y-x)dt \geq \int_0^1 (1-t)\mu\|y-x\|^2 dt = \frac{\mu}{2}\|y-x\|^2.$$

Thus, combining these results gives the definition of μ -strong convexity.

Setting $\mu = 0$ in the above shows the convex case in the same way. □

Quadratic functions whose Hessians are positive definite, indefinite, and negative definite are illustrated in Figure 2. We can confirm the correspondence between positive definiteness and convexity.

Positive definite

Indefinite

Negative definite

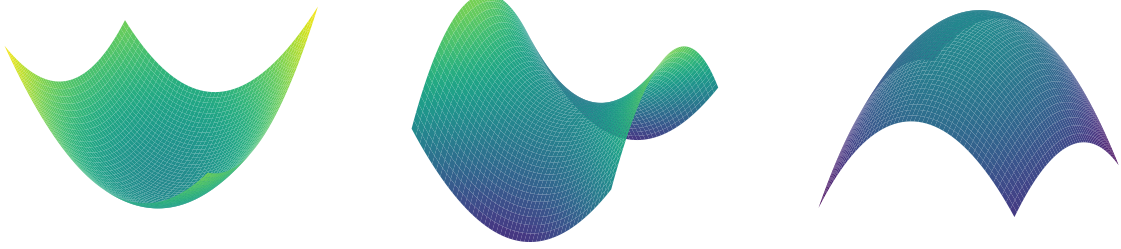


Figure 2: Quadratic function $f(x) = \frac{1}{2}(x - x_k)^\top H(x - x_k) + \nabla f(x_k)^\top (x - x_k) + f(x_k)$ in 2-dimensional space with Hessians H which is (left) positive definite, (center) indefinite, (right) negative definite. $f(x)$ is (left) convex, (center) nonconvex, and (right) concave, respectively.

1.3 Smoothness

Finally, we introduce L -smoothness of functions. A function f is L -smooth if and only if there exists a constant $L > 0$ such that

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|$$

holds for all x, y . The L -smoothness of f is equivalent to the L -Lipschitz continuity of the gradient ∇f by definition. Please also refer to Figure 3.

In general, for nonconvex functions, the following upper bound holds.

Proposition 3. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be an L -smooth function. Then for any $x, y \in \mathbb{R}^n$, we have*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

Proof. By the fundamental theorem of calculus, for any $x, y \in \mathbb{R}^n$, letting $\phi(t) := f(x + t(y - x))$ gives

$$\begin{aligned} f(y) - f(x) &= \phi(1) - \phi(0) = \int_0^1 \phi'(t) dt \\ &= \int_0^1 \nabla f(x + t(y - x))^\top (y - x) dt \\ &= \nabla f(x)^\top (y - x) + \int_0^1 (\nabla f(x + t(y - x)) - \nabla f(x))^\top (y - x) dt \\ &\leq \nabla f(x)^\top (y - x) + \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \|y - x\| dt && \text{(Cauchy-Schwarz inequality)} \\ &\leq \nabla f(x)^\top (y - x) + \int_0^1 Lt \|y - x\|^2 dt && (L\text{-smoothness}) \\ &= \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2. \end{aligned}$$

This proves the upper bound. □

If f is convex in addition, the following lower bound also holds.

Proposition 4. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex and L -smooth function. Then for any $x, y \in \mathbb{R}^n$, we have*

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2.$$

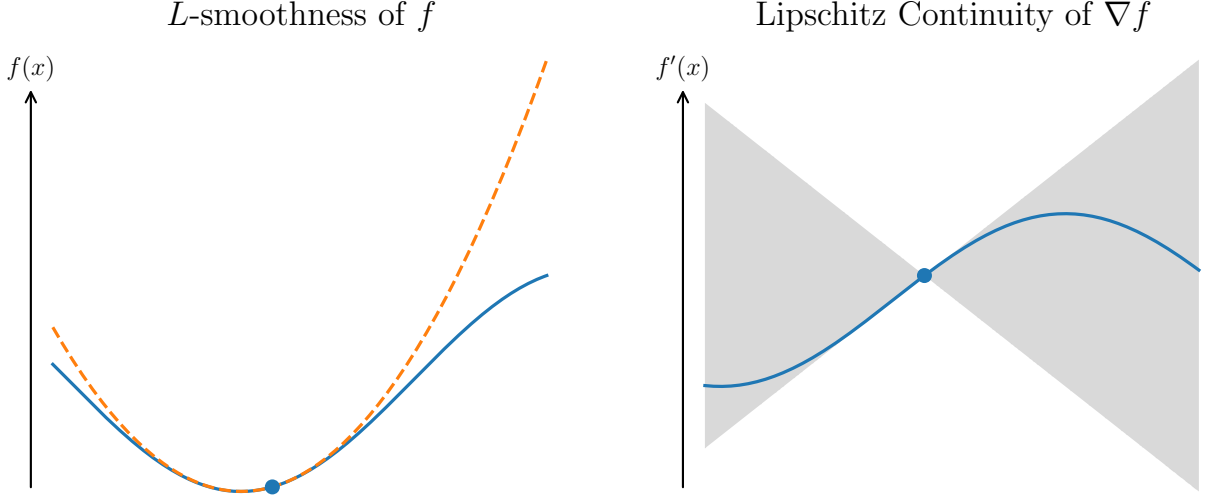


Figure 3: An example of L -smooth function in one-dimensional space. L -smoothness gives an upper bound of $f(x)$ as shown by the orange dashed line, and it is equivalent to the Lipschitz continuity of ∇f , which means that ∇f only exists in the gray area.

Proof. Let us define the function $g_x(z)$ as follows.

$$\begin{aligned} g_x(z) &:= f(z) - f(x) - \nabla f(x)^\top (z - x), \\ \nabla g_x(z) &= \nabla f(z) - \nabla f(x). \end{aligned}$$

Since both $f(z)$ and the linear function $-f(x) - \nabla f(x)^\top (z - x)$ are convex in z , $g_x(z)$ is also convex and attains its minimum at $z = x$ where $\nabla g_x(z) = 0$ and thus the minimum value is $g_x(x) = 0$. Moreover, the L -smoothness of $f(z)$ directly implies the L -smoothness of $g_x(z)$. Thus, by the L -smoothness of $g_x(z)$, we have

$$\begin{aligned} 0 &\leq g_x\left(y - \frac{\nabla g_x(y)}{L}\right) && \text{(minimum is } g_x(x) = 0\text{)} \\ &\leq g_x(y) - (\nabla g_x(y))^\top \frac{\nabla g_x(y)}{L} + \frac{L}{2} \left\| \frac{\nabla g_x(y)}{L} \right\|^2 && \text{(by } L\text{-smoothness)} \\ &= g_x(y) - \frac{1}{2L} \|\nabla g_x(y)\|^2 \\ &= f(y) - f(x) - \nabla f(x)^\top (y - x) - \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2. && \text{(definition of } g_x\text{)} \end{aligned}$$

This proves the lower bound. $\square$

As we have seen so far, L -smoothness means that the rate of gradient change is bounded from above, and this can also be related to the definiteness of the Hessian matrix. In other words, as shown in the following proposition, the Hessian matrix is bounded by $-LI$ and $+LI$.

Proposition 5. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^2 . Then f is L -smooth if and only if $-LI \preceq \nabla^2 f(x) \preceq +LI$ holds for all $x \in \mathbb{R}^n$.*

Proof. Since f is of class C^2 , by the fundamental theorem of calculus, for any $x, y \in \mathbb{R}^n$, we have

$$\nabla f(y) - \nabla f(x) = \int_0^1 \nabla^2 f(x + t(y - x))(y - x) dt.$$

Assume $-LI \preceq \nabla^2 f(x) \preceq +LI$ for all $x \in \mathbb{R}^n$. It implies that the operator norm of $\nabla^2 f(x)$ satisfies $\|\nabla^2 f(x)\| \leq L$, and thus

$$\|\nabla f(y) - \nabla f(x)\| = \left\| \int_0^1 \nabla^2 f(x + t(y - x))(y - x) dt \right\| \quad \text{(previous equation)}$$

$$\begin{aligned}
&\leq \int_0^1 \|\nabla^2 f(x+t(y-x))(y-x)\| dt && \text{(triangle inequality)} \\
&\leq \int_0^1 L \|y-x\| dt && \text{(by assumption)} \\
&= L \|y-x\|,
\end{aligned}$$

which proves the L -smoothness of f .

Conversely, if f is L -smooth, then for any $x \in \mathbb{R}^n$ and $v \in \mathbb{R}^n$, we have

$$\|\nabla f(x+tv) - \nabla f(x)\| \leq L \|tv\| = Lt \|v\|.$$

By Taylor's theorem, with remainder $r(t)$ satisfies $\|r(t)\|/t \rightarrow 0$ as $t \rightarrow 0$, we also have

$$\nabla f(x+tv) - \nabla f(x) = t\nabla^2 f(x)v + r(t).$$

Rearranging this gives

$$\nabla^2 f(x)v = \lim_{t \rightarrow 0} \frac{\nabla f(x+tv) - \nabla f(x) - r(t)}{t} = \lim_{t \rightarrow 0} \frac{\nabla f(x+tv) - \nabla f(x)}{t}.$$

Taking the inner product with v gives

$$\begin{aligned}
v^\top \nabla^2 f(x)v &= \lim_{t \rightarrow 0} \left(\frac{\nabla f(x+tv) - \nabla f(x)}{t} \right)^\top v \\
&\leq \lim_{t \rightarrow 0} \frac{\|\nabla f(x+tv) - \nabla f(x)\|}{t} \|v\| && \text{(Cauchy-Schwarz inequality)} \\
&\leq \lim_{t \rightarrow 0} \frac{L \|tv\|}{t} \|v\| && \text{(by } L\text{-smoothness definition)} \\
&= L \|v\|^2.
\end{aligned}$$

Because $v \in \mathbb{R}^n$ is arbitrary, we obtain $\nabla^2 f(x) \preceq LI$. In the same manner, $\nabla^2 f(x) \succeq -LI$ can also be shown, which completes the proof. $\square$

1.3.1 Bound on Hessian Eigenvalues

By combining the discussions on L -smoothness and μ -strong convexity so far, we can also obtain bounds on the eigenvalues of the Hessian.

Proposition 6. *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^2 . If f is L -smooth and μ -strongly convex, then the eigenvalues of the Hessian $\nabla^2 f(x)$ are contained in the interval $[\mu, L]$ for all $x \in \mathbb{R}^n$.*

Proof. By Propositions 2 and 5, for all $x \in \mathbb{R}^n$, we have

$$\mu I \preceq \nabla^2 f(x) \preceq LI.$$

This directly implies that the eigenvalues of $\nabla^2 f(x)$ are contained in the interval $[\mu, L]$. $\square$

Thus, we can see that μ -strong convexity and L -smoothness correspond to the lower and upper bounds of the eigenvalues of the Hessian, respectively. The eigenvalues of the Hessian play a significant role in determining the convergence performance of algorithms, and such bounds can be utilized in various convergence analyses.

1.3.2 Baillon-Haddad Theorem

As an advanced topic, we introduce the following Baillon-Haddad theorem, which is one of the useful properties of L -smooth functions. Note that the C^2 assumption is not required in this theorem, and only the C^1 assumption is sufficient.

Proposition 7 (Baillon–Haddad theorem). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^1 . If f is L -smooth and convex, then for all $x, y \in \mathbb{R}^n$, ∇f is $1/L$ -cocoercive, i.e.,*

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2$$

for all $x, y \in \mathbb{R}^n$.

Proof. By the lower bound of f for convex and L -smooth functions shown above, we have

$$\begin{cases} f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2, \\ f(x) \geq f(y) + \nabla f(y)^\top (x - y) + \frac{1}{2L} \|\nabla f(x) - \nabla f(y)\|^2. \end{cases}$$

Adding these two inequalities gives the Baillon–Haddad theorem. □

We also guide readers to other literature for the proof [1] [3, Proposition 12.60].

This theorem is sometimes used to derive the following inequality, which is useful for analyzing the update rules of quasi-Newton methods such as BFGS and L-BFGS methods. Assume that the function f is convex and L -smooth. For the sequences $\{x_k\}_k$ generated by an optimization algorithm, defining

$$s_k := x_{k+1} - x_k, \quad y_k := \nabla f(x_{k+1}) - \nabla f(x_k)$$

and applying Baillon–Haddad theorem with $x = x_{k+1}$ and $y = x_k$ yields

$$s_k^\top y_k \geq \frac{1}{L} \|y_k\|^2.$$

This inequality is sometimes used as one of the conditions for ensuring the positive definiteness of the approximate Hessian in quasi-Newton methods, and it plays an important role in theoretical guarantees.

References

- [1] Heinz H. Bauschke and Patrick L. Combettes. *The Baillon-Haddad Theorem Revisited*, 2009.
- [2] Yurii Nesterov. *Introductory Lectures on Convex Optimization: A Basic Course*. Springer Publishing Company, Incorporated, 1 edition, 2014. ISBN 978-1-4613-4691-3.
- [3] R. Tyrrell Rockafellar and Roger J. B. Wets. *Variational Analysis*, volume 317 of *Grundlehren Der Mathematischen Wissenschaften*. Springer, 1998. ISBN 978-3-540-62772-2 978-3-642-02431-3. doi: 10.1007/978-3-642-02431-3.